

# Gaussian Head Avatar: Ultra High-fidelity Head Avatar via Dynamic Gaussians

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Figure 1. Gaussian head avatar achieves ultra high-fidelity image synthesis with controllable expressions at 2K resolution. The above shows different views of the synthesized avatar, and the bottom shows different identities animated by the same expression. 16 views are used during the training.

## Abstract

Creating high-fidelity 3D head avatars has always been a research hotspot, but there remains a great challenge under lightweight sparse view setups. In this paper, we propose Gaussian Head Avatar represented by controllable 3D Gaussians for high-fidelity head avatar modeling. We optimize the neutral 3D Gaussians and a fully learned MLP-based deformation field to capture complex expressions. The two parts benefit each other, thereby our method can model fine-grained dynamic details while ensuring expression accuracy. Furthermore, we devise a well-designed geometry-guided initialization strategy based on implicit SDF and Deep Marching Tetrahedra for the stability and convergence of the training procedure. Experiments show our approach outperforms other state-of-the-art sparse-view methods, achieving ultra high-fidelity rendering quality at 2K resolution even under exaggerated expressions. Project page: <https://yuelangx.github.io/gaussianheadavatar>.

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## 1. Introduction

High-fidelity 3D human head avatar modeling is of great significance in many fields, such as VR/AR, telepresence, digital human and film production. Automatically creating high-fidelity avatars has been a research hotspot in computer vision for decades. Although some traditional head avatars [37, 39, 41, 54] can realize high-fidelity animation, they typically require accurate geometries reconstructed and tracked from dense multi-view videos, thus limiting their applications in lightweight settings.

Thanks to the Neural Radiance Fields (NeRF) [43] which show great capability of novel view synthesis in the absence of accurate geometry, recent methods [38, 57] skip the geometry reconstruction and tracking steps but directly learn high-quality NeRF-based head avatars. Other works [42, 48, 71] have verified that NeRF can be applied to either dense or sparse views, which greatly lowers the threshold for head avatar reconstruction. However, it still remains challenging for these NeRF-based approaches to synthesize high-fidelity images at 2K resolutions with pixel-

level details, including wrinkles and eyes.

To overcome this bottleneck and further improve the avatar quality, we introduce 3D Gaussian splatting [26] for 3D head avatar modeling. This is an explicit discrete representation that can be well adapted to the rasterization-based rendering pipeline. It has been verified that the 3D Gaussian representation is capable of rendering complex scenes with low computational consumption. Compared to NeRF, the reconstruction quality of static and dynamic scenes [40, 59, 67] is much better while rendering time cost has been significantly reduced. Motivated by this progress, we propose Gaussian Head Avatar, a novel representation that utilizes 3D Gaussian splatting for ultra high-fidelity head avatar modeling. Although recent 4D Gaussian works [40, 59, 66, 67] have been proposed to reconstruct dynamic scenes, all of them cannot be animated. For modeling the animatable head avatar, it is crucial but still unexplored how to effectively control the deformation of 3D Gaussians and model the dynamic appearances through expression coefficients.

Previous explicit [74] and implicit [2, 73, 76] head avatars usually formulate the facial deformation via linear blend skinning (LBS) using the skinning weights and blendshapes like the FLAME model [32]. However, such an LBS-based formulation fails to represent exaggerated and fine-grained expressions by simple linear operations, limiting the representation ability of the head avatars. Inspired by NeRsemble [28], we propose a fully learnable expression-conditioned deformation field for the 3D head Gaussians, avoiding the limited capability of the LBS-based formulation. Specifically, we input the positions of the 3D Gaussians with expression coefficients into an MLP to directly predict the displacements from the neutral expression to the target one. Similarly, we control the motion of non-face areas, such as the neck, using the head pose as the condition. 3D Gaussian-based representation has the powerful ability to reconstruct high-frequency details, enabling our method to learn accurate deformation fields. In turn, the learned accurate deformation field facilitates the dynamic Gaussian head model to fit more dynamic details. As a result, our method is able to reconstruct finer-grained dynamic details of expressive human heads.

Unfortunately, as a discretized representation, the gradients back-propagated to the 3D Gaussians cannot spread through the whole space. Thus the convergence of training heavily relies on a plausible initialization for both the geometry and the deformation field. However, simply initializing the 3D head Gaussians with a morphable template like FLAME [32] fails to model the long hairstyle and the shoulders. Hence, we further propose an efficient and well-designed geometry-guided initialization strategy. Specifically, instead of starting from stochastic Gaussians or a FLAME model, we initially optimize an implicit signed dis-

tance function (SDF) field along with a color field and a deformation MLP for modeling the basic geometry, color, and the expression-conditioned deformations of the head avatar respectively. The SDF field is converted to a mesh through Deep Marching Tetrahedra (DMTet) [52], with the color and deformation of the vertices predicted by the MLPs. Then we render the mesh and optimize them jointly under the supervision of multi-view RGB images. Finally, we use the mesh with per-vertex features from the SDF field to initialize the 3D Gaussians to lie on the basic head surface while the color and deformation MLPs are carried over to the next stage, ensuring stable training for convergence. The entire initialization process takes only around 10 minutes.

The contributions of our method can be summarized as:

- We propose Gaussian Head Avatar, a new head avatar representation that employs controllable dynamic 3D Gaussians to model expressive human head avatars, producing ultra high-fidelity synthesized images at 2K resolutions.
- For modeling high-frequency dynamic details, we employ a fully learned deformation field upon the 3D head Gaussians, which accurately model extremely complex and exaggerated facial expressions.
- We carefully design an efficient initialization strategy that leverages implicit representations to initialize the geometry and deformation, leading to efficient and robust convergence when training the Gaussian Head Avatar.

Benefiting from these contributions, our method surpasses recent state-of-the-art methods under lightweight sparse-view setups on the avatar quality by a large margin.

## 2. Related Works

**3D Head Avatar Reconstruction.** Due to the wide application value in the film and digital human industry, 3D head avatar reconstruction from multi-view images has always been a research hotspot. Traditional works [4, 8, 19, 31] reconstruct the scan geometry through multi-view stereo and then register a face mesh template to it. However, such methods usually require heavy computation. With the utilization of deep neural networks, current methods [7, 33, 61, 65] achieve very fast reconstruction, producing even more accurate geometry. Lombardi et al. [37], Bi et al. [5] and Ma et al. [41] represent the full head mesh through a deep neural network and train it with multi-view videos as supervision. However, due to the errors in geometric estimation, mesh-based head avatars typically suffer from texture blur. Therefore, some recent methods [38, 57] utilize NeRF representation [43] to synthesize novel view images without geometry reconstruction, or build NeRF on the head mesh template [39]. Furthermore, the NeRF-based methods are extended to sparse view reconstruction tasks [28, 42, 48, 71] and achieve impressive performance.

Methods which focus on generative model [6, 9, 11, 32, 46, 56, 60] are dedicated to learning general mesh

face templates from large-scale multi-view face images or 3D scans. Recently, implicit SDF-based [68] or NeRF-based [14, 23, 53, 55, 75] methods can learn full-head templates without the limitations of fixed topology, thereby better modeling complex hairstyles and glasses. Cao et al. [14] adopts a hybrid representation of local NeRF built on the mesh surface, which enables high-fidelity rendering and flexible expression control.

3D head avatars reconstruction from monocular videos is also a popular yet challenging research topic. Early methods [12, 13, 15, 24, 25, 45] optimize a morphable mesh to fit the training video. Recent methods [20, 27] leverage neural networks to learn non-rigid deformation upon 3DMM face templates [18, 32], thus can recover more dynamic details. Such methods are not flexible enough to handle complex topologies. Therefore, the latest methods explore to construct head avatar models based on implicit SDF [73], point clouds [74] or NeRF [2, 3, 16, 17, 21, 36, 47, 63, 64, 76].

**Point-based Rendering.** Point elements as a discrete and unstructured representation can fit geometry with arbitrary topology [69] efficiently. Recent methods [29, 30, 58] open up a differentiable rasterization pipeline, such that the point-based representation is widely used in multi-view reconstruction tasks. Aliev et al. [1] and Ruckert et al. [49] propose to first render the feature map, which is transferred to the images through a convolutional renderer. Xu et al. [62] use neural point cloud associated with neural features to model a NeRF. Recently, 3D Gaussian splatting [26] shows its superior performance, beating NeRF in both novel view synthesis quality and rendering speed. Some approaches [34, 40, 50, 59, 66, 67, 72, 72] extend Gaussian representation to dynamic scene reconstruction. However, these methods can not be migrated to the head avatar reconstruction tasks.

### 3. Overview

The pipeline of the reconstruction of Gaussian Head Avatar is illustrated in Fig. 2, including the initialization stage and the training stage of Gaussian Head Avatar. Before the beginning of the pipeline, we remove the background [35] of each image and jointly estimate the 3DMM model [18], 3D facial landmarks and the expression coefficients for each frame. In the initialization stage (Sec. 4.3), we reconstruct an SDF-based neutral geometry, and optimize a deformation MLP and a color MLP from the training data as the guidance model. Next, we extract the neutral mesh through DMTet to initialize the neutral Gaussians while the deformation and color MLPs are also inherited from the initialization stage. In the training stage of Gaussian Head Avatar (Sec. 4.2), we deform the neutral Gaussians to the target expression through the dynamic generator given the driving expression coefficients as the condition. Finally, given a camera view, the expressive Gaussians are rendered to a

feature map, which is then fed into the convolutional super resolution network to generate high-resolution avatar images. The whole model is optimized under the supervision of multi-view RGB videos.

## 4. Method

### 4.1. Avatar Representation

Generally, the static 3D Gaussians [26] with  $N$  points are represented by their positions  $X$ , the multi-channel color  $C$ , the rotation  $Q$ , scale  $S$  and opacity  $A$ . The rotation  $Q$  is represented in the form of quaternion. Subsequently, the Gaussians can be rasterized and rendered to the multi-channel image  $I$  given the camera parameters  $\mu$ . This process can be formulated as:

$$I = \mathcal{R}(X, C, Q, S, A; \mu). \quad (1)$$

Our task is to reconstruct a dynamic head avatar controlled by expression coefficients. Therefore, we formulate the head avatar as dynamic 3D Gaussians conditioned on expressions. To handle the dynamic changes, we input the expression coefficients with head pose to the head avatar model and output the position and other attributes of the Gaussians as above.

Specifically, we first construct a canonical neutral Gaussian model with expression-independent attributes:  $\{X_0, F_0, Q_0, S_0, A_0\}$ , which are fully optimizable.  $X_0 \in \mathbb{R}^{N \times 3}$  denotes the positions of the Gaussians with a neutral expression in the canonical space.  $F_0 \in \mathbb{R}^{N \times 128}$  denotes the point-wise feature vectors as their intrinsic properties.  $Q_0 \in \mathbb{R}^{N \times 4}$ ,  $S_0 \in \mathbb{R}^{N \times 3}$  and  $A_0 \in \mathbb{R}^{N \times 1}$  denotes the neutral rotation, scale and opacity respectively. Note that we do not define the neutral color, but directly predict expression-dependent dynamic color from the point-wise feature vectors  $F_0$ . Then, we construct an MLP-based expression conditioned dynamic generator  $\Phi$  to generate all the extra dynamic changes to the neutral model. Overall, the whole Gaussian head avatar can be formulated as:

$$\{X, C, Q, S, A\} = \Phi(X_0, F_0, Q_0, S_0, A_0; \theta, \beta), \quad (2)$$

with  $\theta$  denoting expression coefficients and  $\beta$  denoting the head pose. During the training, we optimize all the parameters of the dynamic generator  $\Phi$  and the neutral Gaussian model  $\{X_0, F_0, Q_0, S_0, A_0\}$ , which are highlighted in bold in the following.

Next, we explain the process of adding expression-related changes to the neutral Gaussian model through the dynamic generator  $\Phi$  as described in Eqn. 2 in detail.

**Positions**  $X'$  of the Gaussians. Expressions bring about the geometric deformation of the neutral model, which is modeled as the displacements of the Gaussian points.



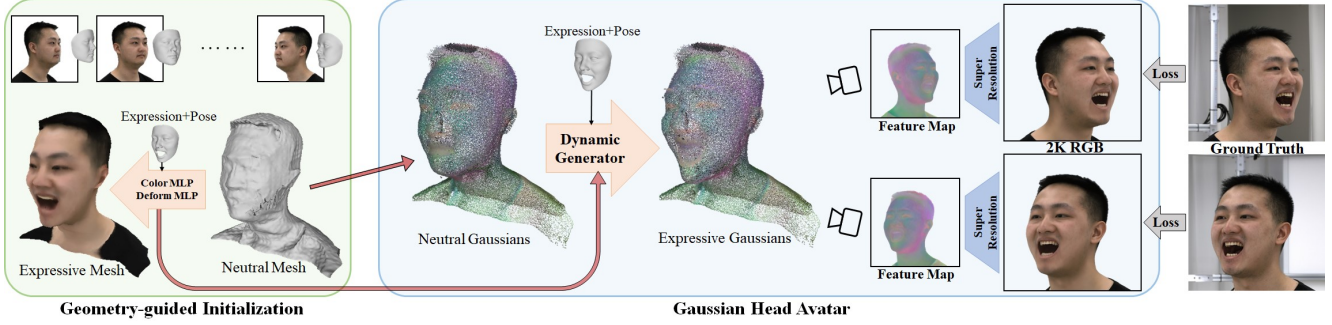


Figure 2. The overview of the Gaussian Head Avatar rendering and reconstruction. We first optimize the guidance model including a neutral mesh, a deformation MLP and a color MLP in the Initialization stage. Then we use them to initialize the neutral Gaussians and the dynamic generator. Finally, 2K RGB images are synthesized through differentiable rendering and the super-resolution network. The Gaussian Head Avatar are trained under the supervision of multi-view RGB videos.

Specifically, we predict the displacements respectively controlled by the expression and the head pose in the canonical space through two different MLPs:  $\mathbf{f}_{def}^{exp} \in \Phi$  and  $\mathbf{f}_{def}^{pose} \in \Phi$ . Then, we add them to the neutral positions.

$$\begin{aligned} X' &= \mathbf{X}_0 + \lambda_{exp}(\mathbf{X}_0) \mathbf{f}_{def}^{exp}(\mathbf{X}_0, \theta) \\ &\quad + \lambda_{pose}(\mathbf{X}_0) \mathbf{f}_{def}^{pose}(\mathbf{X}_0, \beta). \end{aligned} \quad (3)$$

$\lambda_{exp}(\cdot)$  and  $\lambda_{pose}(\cdot)$  represent the extent to which the points are affected by the expression or the head pose respectively. Without decoupling but just using the expression coefficients as the global condition [16] which also controls the shoulders and upper body, will produce jittering results during the animation. Here, we assume that the Gaussian points closer to 3D landmarks are more affected by the expression coefficients and less affected by the head pose, while the opposite is true for the Gaussian points far away. Specifically, The 3D landmarks  $\mathbf{P}_0$  of the canonical model are first estimated through the 3DMM model in the data preprocessing and then optimized in the initialization stage 4.3. Then for each Gaussian point, we calculate the above weight  $\lambda_{exp}(\cdot)$  and  $\lambda_{pose}(\cdot)$  as follows:

$$\lambda_{exp}(x) = \begin{cases} 1, & dist(x, \mathbf{P}_0) < t_1 \\ \frac{t_2 - dist(x, \mathbf{P}_0)}{t_2 - t_1}, & dist(x, \mathbf{P}_0) \in [t_1, t_2] \\ 0, & dist(x, \mathbf{P}_0) > t_2 \end{cases}$$

with  $\lambda_{pose}(x) = 1 - \lambda_{exp}(x)$ . And  $x \in \mathbf{X}_0$  denotes the position of one neutral Gaussian.  $dist(x, \mathbf{P}_0)$  denotes the minimum distance from the point  $x$  to the 3D landmarks  $\mathbf{P}_0$ .  $t_1 = 0.15$  and  $t_2 = 0.25$  are predefined hyperparameters when the length of the head is set to approximately 1.

**Color**  $C'$  of the Gaussians. Modeling the dynamic details typically requires dynamic color that changes with expressions. As we do not pre-define the neutral value in Eqn. 2, the color are directly predict by two color MLPs:

$$\mathbf{f}_{col}^{exp} \in \Phi \text{ and } \mathbf{f}_{col}^{pose} \in \Phi:$$

$$\begin{aligned} C' &= \lambda_{exp}(\mathbf{X}_0) \mathbf{f}_{col}^{exp}(\mathbf{F}_0, \theta) \\ &\quad + \lambda_{pose}(\mathbf{X}_0) \mathbf{f}_{col}^{pose}(\mathbf{F}_0, \beta). \end{aligned} \quad (4)$$

**Rotation, Scale and Opacity**  $\{Q', S', A'\}$  of the Gaussians. These three attributes also dynamic, thereby modeling some detailed expressions-related appearance changes. Here, we just use another two attribute MLPs  $\mathbf{f}_{att}^{exp} \in \Phi$  and  $\mathbf{f}_{att}^{pose} \in \Phi$  to predict their shift from the neutral value.

$$\begin{aligned} \{Q', S', A'\} &= \{\mathbf{Q}_0, \mathbf{S}_0, \mathbf{A}_0\} \\ &\quad + \lambda_{exp}(\mathbf{X}_0) \mathbf{f}_{att}^{exp}(\mathbf{F}_0, \theta) \\ &\quad + \lambda_{pose}(\mathbf{X}_0) \mathbf{f}_{att}^{pose}(\mathbf{F}_0, \beta). \end{aligned} \quad (5)$$

Finally, we apply rigid rotations and translations  $T(\cdot)$  to the Gaussians, transforming them from the canonical space to the world space. Note, the transformation is only implemented for directional variables:  $\{X', Q'\}$ , while the multi-channel color, the scale and the opacity  $\{C', S', A'\}$  are not directional thus remain unchanged.

$$\{X, Q\} = T(\{X', Q'\}, \beta), \quad (6)$$

$$\{C, S, A\} = \{C', S', A'\}. \quad (7)$$

## 4.2. Training

In this part, we explain the training pipeline of the Gaussian head avatar 4.1 and the loss function. In each iteration, we first generate the expression conditioned 3D Gaussians as Eqn. 2. Then we render a 32-channel image with 512 resolution  $I_C \in \mathbb{R}^{512 \times 512 \times 32}$  referring to Eqn. 1. After that we feed the image to a super resolution network  $\Psi$  to generate a 2048 resolution RGB image  $I_{hr} \in \mathbb{R}^{2048 \times 2048 \times 3}$ , such that more details are recovered and noise caused by uneven ambient light or camera chromatic aberration in the training data will be filtered out [49, 64].

During training, we jointly optimize all the learnable parameters mentioned above in bold, including the neutral

Gaussians:  $\{X_0, F_0, Q_0, S_0, A_0\}$ , the dynamic generator:  $\{f_{col}^{exp}, f_{col}^{pose}, f_{def}^{exp}, f_{def}^{pose}, f_{att}^{exp}, f_{att}^{pose}\}$ , and the super resolution network  $\Psi$ . For the loss function, we only use the foreground RGB images  $I_{gt}$  as supervision to construct an L1 loss and a VGG perceptual loss [70] between the generated images  $I_{hr}$  and the ground truth  $I_{gt}$ . Besides, we encourage the first three channels of the 32-channel feature image  $I_C$  to be RGB channels, which is ensured by a L1 loss term. The total loss is:

$$\mathcal{L} = \|I_{hr} - I_{gt}\|_1 + \lambda_{vgg} VGG(I_{hr}, I_{gt}) + \lambda_{lr} \|I_{lr} - I_{gt}\|_1, \quad (8)$$

with  $I_{lr}$  denoting the first three channels of the 32-channel image  $I_C$ . We set the weights  $\lambda_{vgg} = 0.1$  and  $\lambda_{lr} = 1$ .

### 4.3. Geometry-guided Initialization

Unlike neural networks, the Gaussians act as an unordered and unstructured representation. Random initialization leads to failure to converge while naively using the FLAME model to initialize will significantly reduce the reconstruction quality. In this section, we describe in detail how to optimize a mesh guidance model to provide reliable initialization for the Gaussians in Sec. 4.1.

**Mesh Guidance Model.** Specifically, we first construct a MLP  $f_{sdf}$  to represent a signed distance field. In addition, this network will also output the corresponding feature vector of each point, which is used for predicting the point color. It can be formulated as:

$$s, \eta = f_{sdf}(x), \quad (9)$$

with  $s$  denotes the SDF value,  $\eta$  denotes the feature vector and  $x$  denotes the point position. Then through (DMTet) [51], we can differentially extract a mesh with vertices  $X$ , per-vertex feature vectors  $F$  and its faces. We also predict the per-vertex 32-channel color as Eqn. 4 by the two color MLPs  $f_{col}^{exp}$  and  $f_{col}^{pose}$ . In parallel, we construct the two deformation MLPs:  $f_{def}^{exp}$  and  $f_{def}^{pose}$  as described in Sec. 4.1 to predict the displacements and add them to the vertex positions. This process is similar to Eqn 3 above, with the Gaussian positions  $X_0$  replaced by the vertex positions  $X$ . Finally we also apply rigid rotations and translations to the deformed mesh, transforming it to the world space and render the deformed mesh into an image  $I$  and a mask  $M$  through differentiable rasterization [44] according to the camera parameters  $\mu$ .

**Loss Function and Training.** Next, we can construct the RGB loss and the silhouette loss to train the guidance model:

$$\mathcal{L}_{RGB} = \|I_{r,g,b} - I_{gt}\|_1, \quad (10)$$

$$\mathcal{L}_{sil} = IOU(M, M_{gt}), \quad (11)$$

with  $I_{gt}$  and  $M_{gt}$  denote the ground truth RGB image and mask, respectively.  $IOU(\cdot)$  denotes Intersection over

Union metrics. Note that only the first three channels  $R, G, B$  of the 32-channel image  $I$  are supervised by the ground truth RGB images.

We also use the estimated 3D facial landmarks  $P_{gt}$  as described in Sec. 3 to provide rough guidance for the expression deformation MLP. Specifically, we input the neutral 3D landmarks  $P_0$  into the expression deformation MLP to predict the expression conditioned landmarks  $P$ :

$$P = P_0 + f_{def}^{exp}(P_0, \theta). \quad (12)$$

Then we construct the loss function with 3D facial landmarks  $P_{gt}$  as the supervision:

$$\mathcal{L}_{def} = \|P - P_{gt}\|_2, \quad (13)$$

Besides, we introduce three constraints: (1) a regular term  $\mathcal{L}_{offset}$  to punish all non-zero displacements to prevent the two deformation MLPs from learning a global constant offset [63], (2) a regular term  $\mathcal{L}_{lmk}$  to limit the SDF value at the 3D landmarks to be close to zero, such that the landmarks are located on the surface of the mesh, (3) a Laplacian term  $\mathcal{L}_{lap}$  for maintaining the extracted mesh smooth to a certain extent. Overall, the total loss function is formulated as:

$$\mathcal{L} = \mathcal{L}_{RGB} + \lambda_{sil} \mathcal{L}_{sil} + \lambda_{def} \mathcal{L}_{def} + \lambda_{offset} \mathcal{L}_{offset} + \lambda_{lmk} \mathcal{L}_{lmk} + \lambda_{lap} \mathcal{L}_{lap}, \quad (14)$$

with  $\lambda$  denoting the weights of each term, which are set as follows:  $\lambda_{sil} = 0.1$ ,  $\lambda_{def} = 1$ ,  $\lambda_{offset} = 0.01$ ,  $\lambda_{lmk} = 0.1$  and  $\lambda_{lap} = 100$ . We jointly optimize the MLPs mentioned above with the neutral 3D landmarks  $P_0$  jointly until all MLPs are converged.

**Parameters Transfer.** Finally, we use the roughly trained mesh guidance model to initialize the Gaussian model. Specifically, we extract the neutral mesh with vertices  $X$  and per-vertex features  $F$  through DMTet, and directly assign their values to the neutral positions  $X_0 = X$  and the per-vertex feature vectors  $F_0 = F$  of the neutral Gaussians respectively. Then, we retain all the four optimized MLPs:  $\{f_{col}^{exp}, f_{col}^{pose}, f_{def}^{exp}, f_{def}^{pose}\}$  for the Gaussian model. For the other neutral attributes: rotation, scale and opacity, we adopt the original initialization strategy in Gaussian Splatting [26]. And the parameters of the two attribute MLPs:  $\{f_{att}^{exp}, f_{att}^{pose}\}$  and the super resolution network  $\Psi$  are just randomly initialized.

## 5. Experiments

### 5.1. Implementation Details

In the experiment we use 12 sets of data, 10 of which are from NeRSemble [28], and the other 2 are multi-view video data from HAvatar [71]. For the 10 identities from NeRSemble, each set contains 2500 to 3000 frames, 16

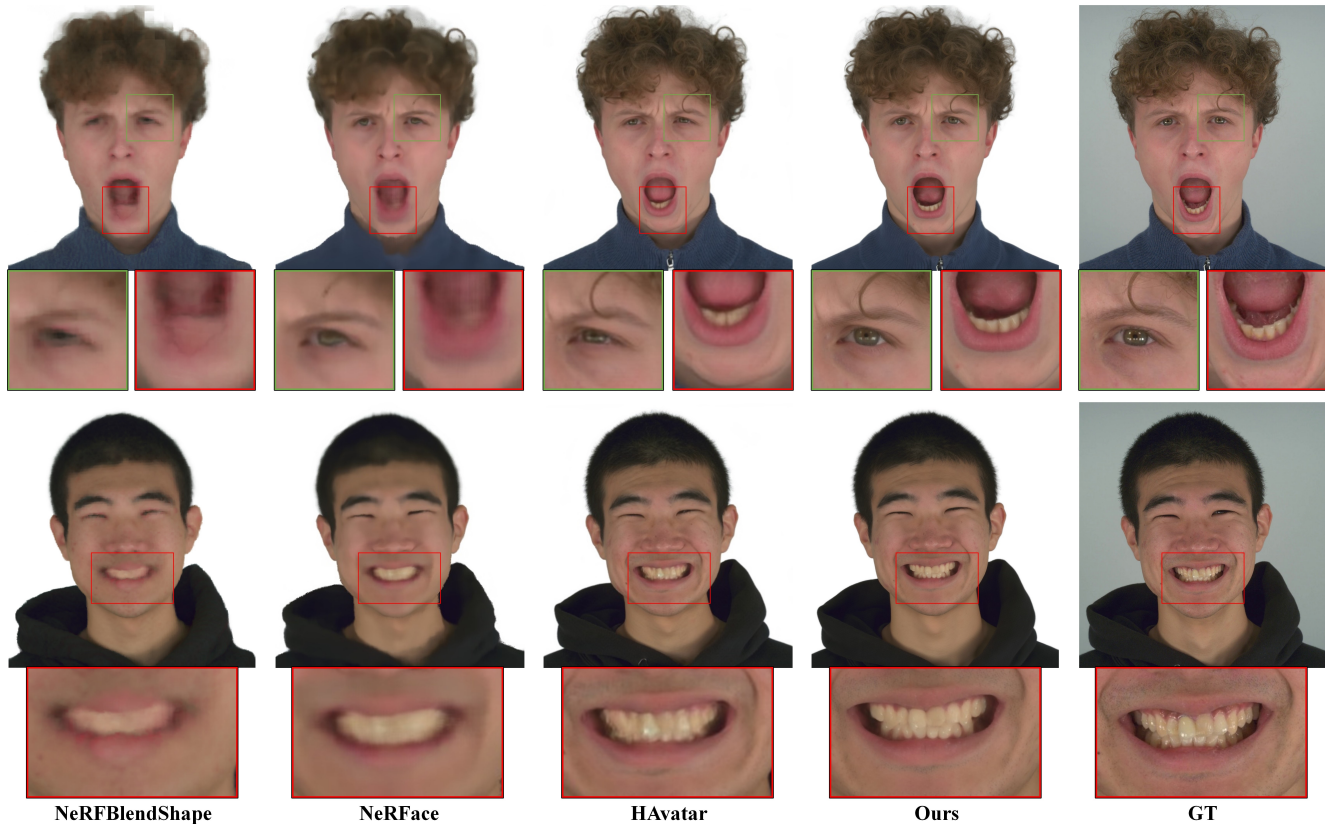


Figure 3. Qualitative comparisons of different methods on self reenactment task. From left to right: NeRFBlendShape [17], NeRFace [16], HAvatar [71] and Ours. Our method can reconstruct details like beards, teeth, etc. with high quality.

cameras are distributed about 120 degrees in front, and simultaneously capture 2K resolution video. For each identity, We use the sequences marked with "FREE" as the evaluation data, and the rest as the training data. For the 2 identities from HAvatar, each set contains 3000 frames, 8 cameras are distributed about 120 degrees in front, and 4K resolution videos are collected simultaneously. Later, we crop the face area and resize to 2K resolution.

For data preprocessing, we first remove the background [35] and extract 68 2D facial landmarks [10] for all the images. Then, for each frame, we use multi-view images to estimate the corresponding 3D landmarks, expression coefficients and head pose by fitting the Basel Face Model (BFM) [18] to the extracted 2D landmarks. Note that we define the 3D landmarks as the usual 68 landmarks with vertices indexed as multiples of 100 in the BFM vertices.

## 5.2. Results and Comparisons

**Self Reenactment.** In this section, we first compare our method with existing SOTA methods in qualitative experiments on self reenactment task. Specifically, NeRFace [16] uses a deep MLP to fit an expression conditioned dynamic NeRF. The current SOTA method HAvatar [71] introduces 3DMM template prior and uses a deep convolutional net-

work to generate a human head NeRF represented by three planes from a mesh template with expression. Note, HAvatar leverages the GAN framework using the adversarial loss function to force the network to generate details that are not view-consistent. For a fair comparison, we remove this part and use VGG perceptual loss as Sec. 4.2 instead.

Qualitative results on self reenactment task are shown in the Fig. 3. Our method can accurately reconstruct pixel-level high-frequency details such as beards, teeth, and hair. Besides, our method can achieve expression transfer more accurately, such as eye movements in the figure.

Next, we conduct a quantitative evaluation for the four methods on 5 identities and 6 cameras using the evaluation split. The evaluation metrics include: Peak Signal-to-Noise Ratio (PSNR), Structure Similarity Index (SSIM), Learned Perceptual Image Patch Similarity (LPIPS) [70] and Fréchet Inception Distance (FID) [22]. Note, we calculate FID by comparing the distribution of all the training images and all the rendered images. As the task mainly focuses on the reconstruction of the head, we use face parsing<sup>1</sup> to remove the body parts in the image to eliminate their impact in the experiment. As shown in Tab. 1, our method demonstrates a slight improvement in PSNR and SSIM compared with

<sup>1</sup><https://github.com/zllrunning/face-parsing.PyTorch>



| Method         | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS (512) $\downarrow$ | LPIPS (2K) $\downarrow$ | FID (2K) $\downarrow$ |
|----------------|-----------------|-----------------|--------------------------|-------------------------|-----------------------|
| NeRFBlendShape | 25.91           | 0.836           | 0.123                    | 0.229                   | 54.80                 |
| NeRFace        | 27.14           | 0.849           | 0.147                    | 0.234                   | 65.11                 |
| HAvatar        | 27.19           | 0.883           | 0.064                    | 0.209                   | 31.06                 |
| Ours (w/o SR)  | <b>27.82</b>    | <b>0.887</b>    | 0.080                    | 0.202                   | 45.50                 |
| Ours           | 27.70           | 0.883           | <b>0.056</b>             | <b>0.098</b>            | <b>18.50</b>          |

Table 1. Quantitative evaluation results of NeRFBlendShape [17], NeRFace [16], HAvatar [71], our method without super resolution and our full method on self reenactment task.

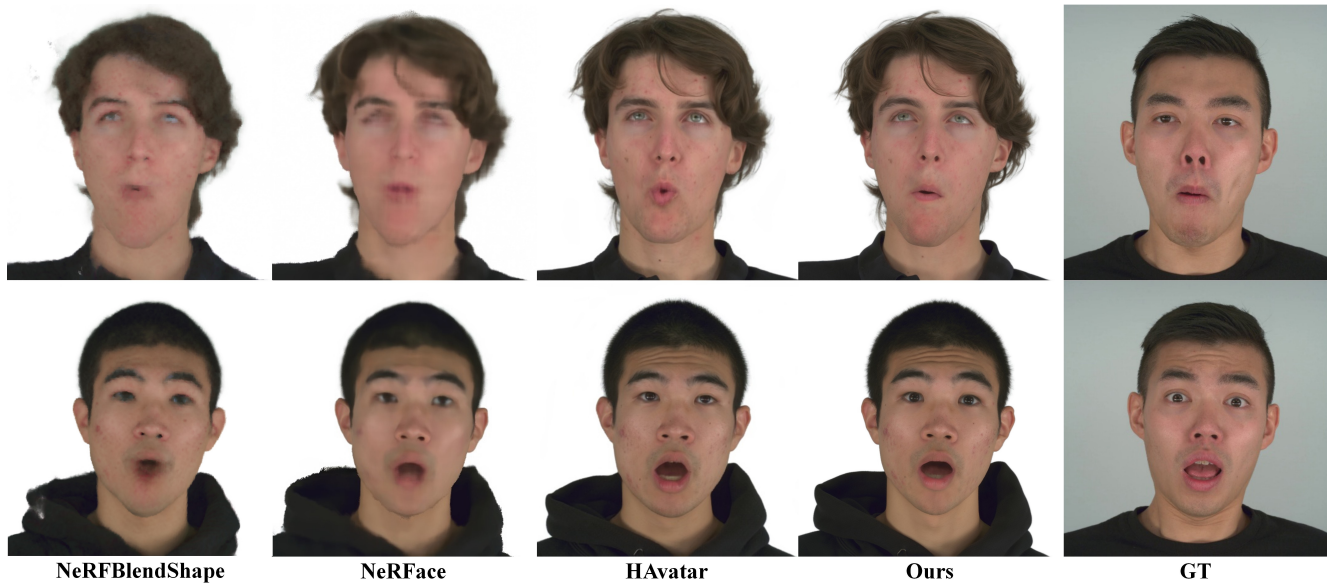


Figure 4. Qualitative comparisons of different methods on cross-identity reenactment task. From left to right: NeRFBlendShape [17], NeRFace [16], HAvatar [71] and Ours. Our method synthesizes high-fidelity images while ensuring the accuracy of expression transfer.

| Method         | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|----------------|-----------------|-----------------|--------------------|
| NeRFBlendShape | 25.43           | 0.812           | 0.148              |
| NeRFace        | 26.65           | 0.825           | 0.151              |
| HAvatar        | 27.13           | 0.880           | 0.65               |
| Ours           | <b>27.58</b>    | <b>0.882</b>    | <b>0.059</b>       |

Table 2. Quantitative evaluation results of the other SOTA methods and our method on 3D consistency.

previous methods, and a significant improvement in LPIPS and FID, which means that our method can generate more high-frequency details.

**Cross-Identity Reenactment.** We also qualitatively compare our method with the above SOTA methods on cross-identity reenactment task. As shown in the results, our method is able to synthesize higher-fidelity images with more accurate expression transfer and richer emotions.

**Novel View Synthesis.** In this section, we show the results of novel view synthesis as shown in Fig. 5 shown. In this case, we use video data from 8 views for training and render the image at a new viewpoint. Next, we quantitatively evaluate the 3D consistency of our method and compare it with other SOTA methods mentioned above. Specif-

ically, we select the 5 identities as above and use the video data from 8 cameras for training while the other 8 hold-out cameras for evaluation. The evaluation metrics include PSNR, SSIM and LPIPS (512 resolution). The results are shown in Tab. 2. Our method outperforms other methods in 3D consistency.

### 5.3. Ablation Study

| Method      | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|-------------|-----------------|-----------------|--------------------|
| FLAME-Init  | 28.73           | 0.875           | 0.123              |
| Mesh-Deform | 28.83           | 0.874           | 0.116              |
| Ours        | <b>28.94</b>    | <b>0.876</b>    | <b>0.108</b>       |

Table 3. Quantitative evaluation results of the other two ablation baselines and ours on self reenactment task.

**Ablation on Initialization Strategies.** In order to verify the effectiveness of our geometry-guided initialization strategy 4.3, we compare it with the strategy to use the FLAME model for initialization (FLAME-Init). Specifically, after fitting a FLAME model through multi-view data, we first subdivide the FLAME mesh 4 times and use the neutral vertices as the positions of the neutral Gaussians. Then, the

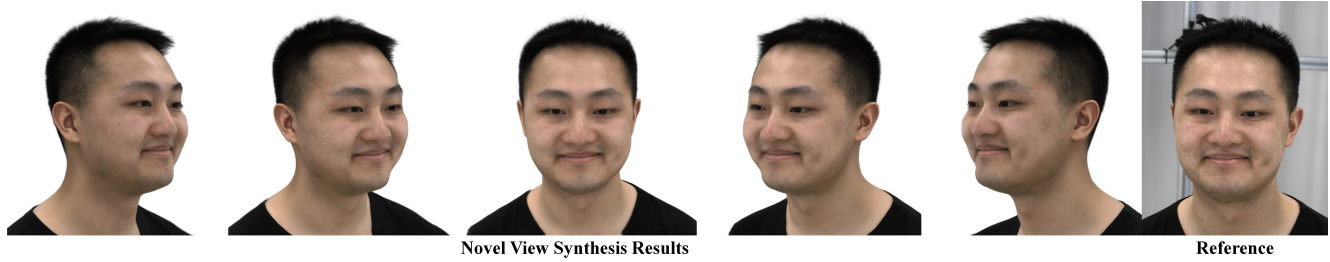


Figure 5. Novel view synthesis results of our method. We use 8-view synchronized videos for training the avatar.

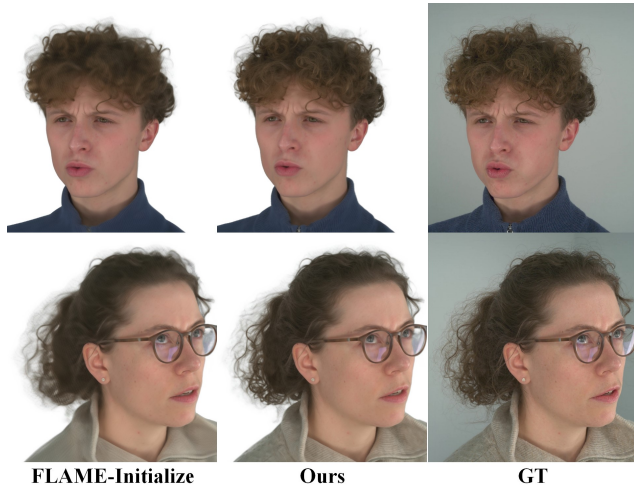


Figure 6. Ablation study on the initialization strategies: FLAME-initialization and our geometry-guided initialization. Our strategy ensures the hair strands away from the head are well reconstructed.

expression deformation MLP is optimized to learn the displacement of FLAME vertices. We set the per-vertex feature to zeros, while randomly initialize the parameters of the expression color MLP. The initialization of other variables is the same as our strategy. Qualitative results are shown in the Fig. 6. Due to the lack of initialization for the hair and shoulders in FLAME-Init, the points to model these parts are offset from nearby vertices, which leads to sparseness of the Gaussians, resulting in blurring.

**Ablation on Deformation Modeling Approaches.** We compare our fully learned deformation field with the previous mesh-based deformation (Mesh-Deform). Specifically, we migrate the method in INSTA [76] for controlling the NeRF deformation to our Gaussians. First we fit a 3DMM mesh template. Then, for each Gaussian point, find the closest face on the mesh, and calculate the deformation gradient to estimate the displacement. Qualitative results are shown in the Fig. 7. For some expressions that cannot be captured well by the 3DMM mesh template, our method can learn accurate deformation, thereby achieving the modeling of complex expressions. Quantitative results are shown in the Fig. 3. Our method outperforms both the two ablation baselines on PSNR, SSIM and LPIPS metrics.

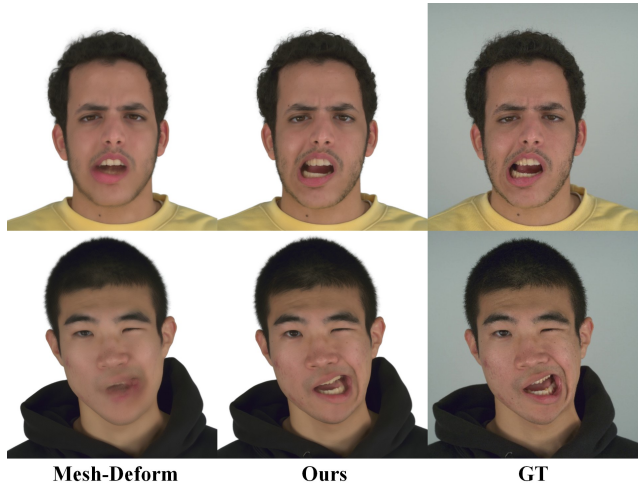


Figure 7. Ablation study on the deformation modeling: mesh LBS-based deformation and our fully learned deformation. Our approach can learn complex and exaggerated expressions.

## 6. Discussion and Conclusion

**Ethical Considerations.** Our method is capable of creating artificial portrait videos, which have the potential to disseminate misinformation, influence public perceptions, and erode confidence in media sources.

**Limitation.** For the tongue and teeth inside the mouth or long hair, blurring is sometimes produced in our method due to the lack of tracking methods.

**Conclusion.** In this paper, we propose Gaussian Head Avatar, a novel representation for head avatar reconstruction, which leverages dynamic 3D Gaussians controlled by a fully learned expression deformation. Experiments demonstrate our method can synthesize ultra high-fidelity images while modeling exaggerated expressions. In addition, we propose a well-designed minute-level initialization strategy to ensure the training convergence. We believe our Gaussian Head Avatar will become the mainstream direction for head avatar reconstruction in the future.

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